AI-Powered Customer Support Ticketing System

Salesforce Service Cloud · Flow Automation · Agentforce

Project documentation and verification summary

Telecommunications support scenarios: broadband outages, eSIM activation, roaming billing, and network incidents.

Scope: Salesforce DX metadata, custom schema, record-triggered Flows, email alerts, support profiles, reports, and Agentforce configuration.

Verification note: Live-org results are based on the supplied implementation report. Salesforce CLI deployment validation has not been performed.

1. Executive Summary

This project configures a Salesforce support workflow for common telecommunications service incidents. It combines Case data, priority-based SLA targets, record-triggered automation, email alerts, support access profiles, operational reports, and an Agentforce assistant configuration.

Implementation highlights

• Custom objects: `Support_Category__c`, `SLA_Policy__c`, and `AI_Feedback__c`.
• Custom Case fields for issue type, support category, SLA deadline, resolution-time target, and suggested reply.
• Validation rule requiring an SLA deadline for High-priority Cases.
• Before-save and after-save record-triggered Flows.
• Three email templates, related email alerts, Lightning application, support profiles, and two reports.

Configured SLA targets

Priority	Target resolution time
High	4 hours
Medium	12 hours
Low	24 hours

The before-save Flow is configured to populate the resolution target, SLA deadline under its conditions, and suggested-reply field. The validation rule remains active. Validate the combined behavior again in any target org before operational use.

2. Architecture and Automation

Custom data model

Object	Purpose
Support_Category__c	Categorizes telecom incidents and stores default priority and active status.
SLA_Policy__c	Stores priority-level resolution targets and escalation settings.
AI_Feedback__c	Stores feedback linked to a Case, including status, suggestion type, and agent comments.
Case (standard)	Extended with issue type, support category, SLA deadline, resolution time, and AI suggested reply.

Record-triggered Flows

Flow	Configured responsibility
Case_Before_Save_SLA_and_Triage_Flow	Before-save field assignments for SLA target/deadline and suggested reply; avoids separate self-record DML.
Case_SLA_and_Triage_Automation_Flow	After-save notification paths for assignment to High priority and transition to Escalated; no self-record update element in the packaged XML.
Case_Resolution_Email_Flow	Resolution notification when a Case transitions to Closed.
Case_Summary_Flow	Autolaunched Flow included as the target of Agentforce-related metadata; successful live invocation was not established by the preview test.

Validation rule

Name: `SLA_Due_Date_Required_For_High_Priority`
Formula: `AND(ISPICKVAL(Priority, "High"), ISBLANK(SLA_Due_Date__c))`
Error: “SLA Due Date must be populated for High priority cases.”

3. Reports, Profiles, and Agentforce

Lightning application and reports

The `AI_Powered_CST_System` Lightning application includes navigation to Cases, Accounts, Contacts, Support Categories, SLA Policies, AI Feedbacks, Reports, and Dashboards. The packaged reports are `Open_Cases_by_Priority` and `SLA_Breach_and_Resolution_Time_Analysis`.

The SLA analysis report includes row-level formula `CDF1` (“SLA Breach Status”). Its packaged formula classifies records as No SLA Target, SLA Breached, Within SLA, or Within SLA (Active), depending on the deadline and status. This is a report formula, not a persistent Case field.

Access control

The package contains Support Agent and Support Manager profiles. Review their object and field permissions against the target org's security model before deployment; profile metadata may interact with existing org configuration.

Agentforce verification result

The live `AI_Customer_Support_Assistant` Version 1 conversation preview was run with a broadband outage scenario. The assistant responded that it could not retrieve the specific troubleshooting guidance and offered escalation to a live agent.

Interpretation: the preview did execute and return a response, but the intended troubleshooting workflow was not demonstrated. Successful custom action invocation and grounded knowledge retrieval remain unverified. The project should not be described as having fully working end-to-end AI troubleshooting until those paths are tested successfully.

4. Live Development-Org Test Matrix

The following records and results were reported from the live development org. Case numbers and auto-numbered feedback IDs are environment-specific and are not recreated automatically by deploying metadata.

Record	Scenario / state	Reported outcome
Case 00001026	Broadband fiber outage; High; New	Validation negative test and before-save SLA/reply population reported successful; report classified overdue.
Case 00001027	International roaming billing dispute; Medium; Closed	12-hour target populated; resolution Flow reported executed; report classified within SLA.
Case 00001028	5G eSIM activation failure; Medium; Working	12-hour target and suggested reply populated; report classified active/within SLA.
Case 00001029	Regional cell tower outage; High; Escalated	Past SLA deadline retained for breach testing; escalation notification path reported triggered.
AF-00001	Feedback linked to Case 00001026	Helpful; Suggested Reply.
AF-00002	Feedback linked to Case 00001027	Helpful; Case Classification.

The table summarizes the supplied live-org report. It is not an independent replay of those tests by this documentation generator.

5. Repository, Validation, and Deployment

Repository contents

The SFDX source package contains custom object and field metadata, validation rules, Flows, email templates, workflow alerts, profiles, tabs, the Lightning application, reports, Agentforce-related metadata, a package manifest, and this documentation PDF.

Local package checks reported

Check	Status
ZIP integrity	Passed; no archive errors reported.
XML syntax	Passed; 43 XML files parsed in the supplied package report.
Manifest cross-reference	Original package was reported as passing; this refined manifest explicitly lists the 15 custom fields and the Case workflow component.
Credential scan	No credentials reported in the supplied scan; org-specific identifiers were removed from this refined README and PDF.
Salesforce CLI deployment validation	Not performed. The Salesforce CLI was unavailable in the reported offline environment.

Deployment outline

1. Install Salesforce CLI and authenticate to a disposable development/test org.
2. Confirm required Salesforce features, including Einstein/Agentforce, are enabled where applicable.
3. Review profile permissions and org-specific email recipients/settings.
4. Run `sf project deploy validate --manifest manifest/package.xml --target-org <your-org-alias>`.
5. Resolve deployment errors and run functional tests before deploying to any intended environment.

Final status: the package is structurally prepared for review, but no claim of successful Salesforce deployment should be made until CLI validation and post-deployment tests pass. Agentforce's intended troubleshooting behavior remains a known limitation.